

PROJECT DOCUMENTATION PHASE

ComicCraft – AI Comic Story Creator using Gemini Models

1. Documentation Overview

The Project Documentation Phase organizes the technical and functional information of ComicCraft into a clear project record. The documentation describes the application purpose, system architecture, AI models, development modules, API routes, frontend templates, deployment process, and future scope.

2. Project Description

ComicCraft is a web-based application that uses AI to generate personalized comic stories and illustrations from user-provided prompts. The user supplies a story prompt, main character name, setting, tone, and art style. The system generates a panel-by-panel storyline and illustrations, provides a web preview, and supports PDF export.

3. Documentation Objectives

The documentation records the following major areas:

- Project purpose and expected workflow.
- Software architecture and technology stack.
- AI model selection and generation process.
- Core Python modules and their responsibilities.
- FastAPI routes and input structures.
- Frontend templates and dynamic rendering.
- Local deployment and testing information.
- Future enhancement possibilities.

4. Main User Workflow

The documented workflow begins with user input, continues through outline generation, story and dialogue generation, image generation, comic layout construction, web preview, and PDF export.

PROJECT DOCUMENTATION PHASE – SYSTEM AND MODEL DOCUMENTATION

1. System Architecture Documentation

ComicCraft follows a three-part architecture consisting of the frontend, backend, and AI integration layers. The frontend uses HTML/CSS and Jinja2 templates. The backend is implemented using FastAPI. AI integration uses Gemini models for text generation and Diffusers/Stable Diffusion for illustrations.

Layer	Technology / Component	Purpose
Frontend	HTML, CSS, Jinja2	Collect inputs and display generated comic content.
Backend	FastAPI	Handle routes, inputs, generation workflow and responses.
AI Text	Gemini Flash / Gemini Pro	Generate outline, narration and dialogue.
AI Image	Stable Diffusion via Diffusers	Generate comic-style panel illustrations.
Export	FPDF	Build the final comic PDF.

2. AI Model Documentation

Gemini Flash is selected for generating the structured five-panel outline. Gemini Pro is used for detailed narration and character dialogue. Stable Diffusion, using the runwayml/stable-diffusion-v1-5 model through Hugging Face Diffusers, is used for comic-style illustrations.

3. AI Generation Sequence

- User prompt and preferences are received by the application.
- Gemini Flash creates a structured five-panel outline.
- Gemini Pro expands the outline into story narration and dialogue.
- Image prompts are used to generate panel illustrations.
- Generated text and images are combined into a comic layout.

PROJECT DOCUMENTATION PHASE – MODULE AND CODE DOCUMENTATION

1. Core Module Documentation

The project documentation identifies separate modules for AI text generation, image generation, layout construction, and PDF export. Each module has a defined responsibility in the comic-generation pipeline.

Module / Function	Documented Responsibility
gemini_flash.py – generate_outline()	Creates the structured five-panel comic outline.
gemini_pro.py – generate_story()	Generates detailed narration and character dialogue.
image_generator.py – generate_image()	Generates panel illustrations from image prompts.
layout_builder.py – build_comic_layout()	Combines images and story content into panel layout data.
exporters.py – save_pdf()	Creates the final comic PDF using FPDF.

2. FastAPI Backend Documentation

The backend is organized around FastAPI routes. The main workflow route accepts the user's form inputs and coordinates the comic-generation process. A JSON route uses the PromptRequest Pydantic model to receive structured input.

3. Input Data Documentation

The documented PromptRequest fields are:

- prompt – the user's story prompt.
- character – the main character name.
- setting – the story setting.
- tone – the desired story tone.
- style – the requested art style.

4. Processing Documentation

The documented processing flow links the input fields to the AI generation functions, image generation, layout construction, preview rendering, and PDF export.

PROJECT DOCUMENTATION PHASE – API, FRONTEND AND FILE DOCUMENTATION

1. Route Documentation

Route	Documented Function
/	Loads the homepage and comic-generation input form.
/generate	Processes form inputs and generates the comic workflow.
/generate-comic/json	Accepts JSON through PromptRequest and returns comic generation data.
/export-success	Displays the successful export page.
/test-image	Tests image generation from a custom prompt.

2. Frontend Template Documentation

The project contains three main Jinja2 templates. `index.html` provides the homepage and input form. `comic_preview.html` displays generated comic panels and their associated content. `export_success.html` provides the result after successful PDF export.

3. Static and Output Documentation

The documented project structure includes `static/panels` for generated panel images and `static/exports` for generated PDF files. Templates are stored separately from backend Python modules.

4. API Documentation Support

FastAPI provides an interactive `/docs` endpoint during local development. This is documented as a way to explore and test the application's API routes.

5. Environment Configuration

The deployment documentation identifies environment variables for the Gemini API key and Hugging Face API token. These values support the external AI services used by ComicCraft.

PROJECT DOCUMENTATION PHASE – SETUP, DEPLOYMENT AND FUTURE SCOPE

1. Development Setup Documentation

The project documentation specifies Python and a virtual environment for development. Required packages include FastAPI, Uvicorn, Jinja2, Google Generative AI, Diffusers, Transformers, Torch, Pillow, Requests, FPDF, and python-dotenv.

2. Local Server Documentation

The documented local deployment uses Uvicorn with the FastAPI application and reload enabled. The application is accessed at 127.0.0.1:8000, with the /docs route available for API exploration.

3. Documentation Checklist

- ✓ Project purpose and workflow documented.
- ✓ System architecture documented.
- ✓ Gemini and Stable Diffusion model roles documented.
- ✓ Core functions and modules documented.
- ✓ FastAPI routes documented.
- ✓ Input fields and JSON request structure documented.
- ✓ Frontend templates documented.
- ✓ Environment variables and local deployment documented.
- ✓ Future enhancement scope documented.

4. Future Scope Documentation

The project document identifies future enhancements including user profiles, comic libraries, and multi-page story arcs. These possibilities extend the current single-comic-generation version.

5. Documentation Outcome

The Project Documentation Phase provides a consolidated technical record of ComicCraft. It connects the project's architecture, AI models, modules, routes, frontend pages, deployment setup, and future scope into a structured reference that can support development, testing, presentation, and maintenance.

Conclusion

Complete documentation ensures that the ComicCraft workflow and its major software components are clearly described from user input through AI generation, comic preview, and PDF export.